- Severe jaundice (yellow colouration of eyes and skin)
- Chronic inflammation of the liver
- Fibrosis and fatty infiltration of the liver
- Weight loss and muscle wasting
- Necrosis (death of cells)



STAGES OF CIRRHOSIS!

The pathogenesis of alcoholic liver disease progresses in three stages.

Stage 1: Hepatic Steatosis or Fatty Liver: During this stage, the fat infiltrates in to functioning liver cells and causes problems in normal functioning of liver. This stage is reversible if alcohol abuse is stopped.

Stage 2: Alcoholic Hepatitis: This is characterized by enlargement of liver. If alcohol intake is discontinued, hepatitis may resolve or it progresses to cirrhosis.

Stage 3: Alcoholic Cirrhosis: Patient develops further complications of ascites, gastrointestinal bleeding, portal hypertension and hepatic encephalopathy in this stage.

Goals for dietary treatment

- To maintain adequate nutrition
- To prevent breakdown of body protein tissue
- To control ascites
- To prevent progression of the disease
- To prevent the symptoms of hepatic encephalopathy

Dietary principle

A high calorie, high carbohydrate, adequate protein and low fat, low fluid and restricted sodium (in case of ascites), adequate fibre and high vitamins and minerals diet is prescribed for the patients with liver cirrhosis.

Diet therapy for cirrhosis

Energy

Approximately 35-40 kcal/ kg body weight is recommended. Energy intake of 2000 to 3000 kcal per day is given to maintain adequate nutrition and to prevent protein breakdown.

Protein

Protein intake should be adjusted as per the individual requirement. A protein intake of 1g/ kg body weight per day is given to achieve positive nitrogen balance. In case of hepatic coma, protein requirement is reduced to 0.5g/ kg bodyweight per day.

Fat

Steatorrhoea or fatty infiltration of liver may be seen in cirrhosis patient. A low intake of fat in the form of medium chain triglyceride (eg Coconut oil) may prove to be effective in reducing the malabsorption of fat.

Carbohydrate

High carbohydrate (300-400g/day) may be given for its protein sparing effect. This protects and supports liver function. Carbohydrate intake also helps in maintaining the weight of the patient.



MALNUTRITION IN CIRRHOSIS!

Malnutrition is common and predominant in liver diseases. The factors resulting in malnutrition are:

- Decreased food intake because of anorexia, nausea and vomiting, unpalatable diet and the drugs
- Impaired digestion and absorption of nutrients due to pancreatic insufficiency, bile salt deficiency and mucosal defect
- Increased energy requirement
- Insufficient protein synthesis
- Alcohol consumption

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